

常微分方程第五次作业

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4.3.1. 设函数 $y = y(x, \eta)$ 是初值问题

$$\begin{cases} \frac{dy}{dx} = \sin xy, \\ y(0) = \eta, \end{cases}$$

的解. 求证:

$$\frac{\partial y}{\partial \eta}(x, \eta) > 0.$$

解答.

记 $u(x, \eta) = \frac{\partial y}{\partial \eta}(x, \eta)$, 则

$$\begin{cases} u'(x, \eta) = x \cos(xy(x, \eta))u(x, \eta), \\ u(0, \eta) = 1, \end{cases}$$

故

$$\frac{\partial y}{\partial \eta}(x, \eta) = u(x, \eta) = e^{\int_0^x s \cos(sy(s, \eta)) ds} > 0.$$

□

4.3.2. 设函数 $\mathbf{y}(x; x_0, \mathbf{y}_0)$ 是初值问题

$$\begin{cases} \frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y}), \\ \mathbf{y}(x_0) = \mathbf{y}_0, \end{cases}$$

的解, 其中函数 $\mathbf{f}(x, \mathbf{y})$ 关于 $(x, \mathbf{y})$ 是连续可微的. 求证:

$$\frac{\partial \mathbf{y}}{\partial x_0}(x; x_0, \mathbf{y}_0) + \frac{\partial \mathbf{y}}{\partial \mathbf{y}_0}(x; x_0, \mathbf{y}_0) \mathbf{f}(x_0, \mathbf{y}_0) \equiv \mathbf{0}.$$

解答.

由初值问题解的唯一性可得 $\mathbf{y}(x; x_0, \mathbf{y}_0) \equiv \mathbf{y}(x; s, \mathbf{y}(s; x_0, \mathbf{y}_0))$, 两边对 s 求导得

$$\mathbf{0} = \frac{\partial \mathbf{y}}{\partial x_0}(x; s, \mathbf{y}(s; x_0, \mathbf{y}_0)) + \frac{\partial \mathbf{y}}{\partial \mathbf{y}_0}(x; s, \mathbf{y}(s; x_0, \mathbf{y}_0)) \frac{d\mathbf{y}}{ds}(s; x_0, \mathbf{y}_0),$$

代入 $s = x_0$ 即证. □

4.3.4. 求给定微分方程初值问题的解对初值或参数的导数:

$$(1) \begin{cases} y' = 2x + \mu y^2, \\ y(0) = \mu - 1, \end{cases} \text{ 求 } \left. \frac{\partial y}{\partial \mu} \right|_{\mu=0};$$

$$(2) \begin{cases} y'' = \frac{2}{x} - \frac{2}{y}, \\ y(1) = 1, y'(1) = \mu, \end{cases} \text{ 求 } \left. \frac{\partial y}{\partial \mu} \right|_{\mu=1}.$$

解答.

(1) 记 $u(x, \mu) = \frac{\partial y}{\partial \mu}(x, \mu)$, 则

$$\begin{cases} u'(x, \mu) = 2\mu y(x, \mu)u(x, \mu) + y(x, \mu)^2, \\ u(0, \mu) = 1, \end{cases}$$

代入 $\mu = 0$ 可得 $y(x, 0) = x^2 - 1$, 故解得

$$\left. \frac{\partial y}{\partial \mu} \right|_{\mu=0} = u(x, 0) = 1 + \int_0^x (s^2 - 1)^2 ds = \frac{1}{5}x^5 - \frac{2}{3}x^3 + x + 1.$$

(2) 记 $u(x, \mu) = \frac{\partial y}{\partial \mu}(x, \mu)$, 则

$$u''(x, \mu) = \frac{2}{y(x, \mu)^2} u(x, \mu), \quad u(1, \mu) = 0, \quad u'(1, \mu) = 1,$$

代入 $\mu = 1$ 可得 $y(x, 1) = x$, 故

$$u''(x, 1) = \frac{2}{x^2} u(x, 1), \quad u(1, 1) = 0, \quad u'(1, 1) = 1,$$

解得

$$\left. \frac{\partial y}{\partial \mu} \right|_{\mu=1} = u(x, 1) = \frac{1}{3}x^2 - \frac{1}{3x}.$$

□

5.1.1. 设 $\Phi(x)$ 是齐次方程的基本解矩阵, 并且函数 $\mathbf{f}(x, \mathbf{y})$ 在区域 $E: x \in (a, b), \|\mathbf{y}\| < +\infty$ 上连续. 求证:

初值问题

$$\begin{cases} \mathbf{y}' = \mathbf{A}(x)\mathbf{y} + \mathbf{f}(x, \mathbf{y}), \\ \mathbf{y}(x_0) = \mathbf{y}_0 \end{cases}$$

等价于积分方程

$$\mathbf{y}(x) = \Phi(x) \left[\Phi^{-1}(x_0)\mathbf{y}_0 + \int_{x_0}^x \Phi^{-1}(s)\mathbf{f}(s, \mathbf{y}(s)) \, ds \right],$$

其中 $x_0 \in (a, b)$.

解答.

(1) 若 $\mathbf{y}(x)$ 满足初值问题, 设 $\mathbf{y}(x) = \Phi(x)\mathbf{c}(x)$, 则 $\mathbf{c}(x_0) = \Phi^{-1}(x_0)\mathbf{y}_0$, 且

$$\begin{aligned} \Phi'(x)\mathbf{c}(x) + \Phi(x)\mathbf{c}'(x) &= \mathbf{y}'(x) \\ &= \mathbf{A}(x)\mathbf{y}(x) + \mathbf{f}(x, \mathbf{y}(x)) \\ &= \mathbf{A}(x)\Phi(x)\mathbf{c}(x) + \mathbf{f}(x, \mathbf{y}(x)) \\ &= \Phi'(x)\mathbf{c}(x) + \mathbf{f}(x, \mathbf{y}(x)), \end{aligned}$$

故

$$\mathbf{c}'(x) = \Phi^{-1}(x)\mathbf{f}(x, \mathbf{y}(x)),$$

从而

$$\mathbf{c}(x) = \Phi^{-1}(x_0)\mathbf{y}_0 + \int_{x_0}^x \Phi^{-1}(s)\mathbf{f}(s, \mathbf{y}(s)) \, ds,$$

因此

$$\mathbf{y}(x) = \Phi(x)\mathbf{c}(x) = \Phi(x) \left[\Phi^{-1}(x_0)\mathbf{y}_0 + \int_{x_0}^x \Phi^{-1}(s)\mathbf{f}(s, \mathbf{y}(s)) \, ds \right],$$

满足积分方程.

(2) 若 $\mathbf{y}(x)$ 满足积分方程, 则

$$\begin{aligned} \mathbf{y}'(x) &= \Phi'(x) \left[\Phi^{-1}(x_0)\mathbf{y}_0 + \int_{x_0}^x \Phi^{-1}(s)\mathbf{f}(s, \mathbf{y}(s)) \, ds \right] + \Phi(x)\Phi^{-1}(x)\mathbf{f}(x, \mathbf{y}) \\ &= \mathbf{A}(x)\Phi(x) \left[\Phi^{-1}(x_0)\mathbf{y}_0 + \int_{x_0}^x \Phi^{-1}(s)\mathbf{f}(s, \mathbf{y}(s)) \, ds \right] + \mathbf{f}(x, \mathbf{y}) \\ &= \mathbf{A}(x)\mathbf{y} + \mathbf{f}(x, \mathbf{y}), \end{aligned}$$

且 $y(x_0) = y_0$, 满足初值问题. □

5.1.2. 设当 $x \in (a, b)$ 时, 非齐次方程中的函数 $f(x)$ 不恒为零, 求证: 该方程有且至多有 $n + 1$ 个线性无关解.

解答.

由于非齐次方程的通解为

$$y(x) = \Phi(x)c + \phi_*(x), \quad c \in \mathbb{R}^n,$$

其中 $\Phi(x)$ 为对应的齐次方程的基本解矩阵, $\phi_*(x)$ 为非齐次方程的一个特解, 且 $\phi_*(x)$ 不能表示为齐次方程的解的线性组合 (否则 $f(x) \equiv 0$), 故该方程有 $n + 1$ 个线性无关解

$$\Phi(x)\varepsilon_1 + \phi_*(x), \quad \Phi(x)\varepsilon_2 + \phi_*(x), \quad \dots, \quad \Phi(x)\varepsilon_n + \phi_*(x), \quad \phi_*(x).$$

同时又由于

$$y(x) = (\Phi(x), \phi_*(x)) \begin{pmatrix} c \\ 1 \end{pmatrix} = (\Phi(x), \phi_*(x))\tilde{c}, \quad \tilde{c} := \begin{pmatrix} c \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1},$$

从而该方程至多有 $n + 1$ 个线性无关解. □

5.1.3. 求证: 向量组

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} x^2 \\ 0 \\ 0 \end{pmatrix}$$

不可能同时满足任何一个三阶齐次方程.

解答.

假设该向量组同时满足一个三阶齐次方程, 则显然它们线性无关, 但 Wronsky 行列式恒为零, 矛盾. □

5.1.4. 求解微分方程组

$$\frac{dx}{dt} = \frac{2}{t}x + 1, \quad \frac{dy}{dt} = \frac{1}{t}x + y.$$

解答.

逐一求解可得

$$x = C_1 t^2 - t, \quad y = C_2 e^t - C_1(t+1) + 1, \quad C_1, C_2 \in \mathbb{R}.$$

□

5.1.5. 考虑初值问题

$$\begin{cases} \frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y}), \\ \mathbf{y}(x_0) = \mathbf{y}_0, \end{cases}$$

其中 $\mathbf{f}(x, \mathbf{y})$ 关于 x 连续, 关于 $\mathbf{y}$ 连续可微. 设 $\mathbf{y} = \phi(x; x_0, \mathbf{y}_0)$ 是其唯一解. 求证:

$$\det \frac{\partial \phi}{\partial \mathbf{y}_0}(x; x_0, \mathbf{y}_0) = e^{\int_{x_0}^x \text{tr}\left(\frac{\partial \mathbf{f}}{\partial \mathbf{y}}(s, \phi(s; x_0, \mathbf{y}_0))\right) ds}.$$

解答.

记 $U(x; x_0, \mathbf{y}_0) = \frac{\partial \phi}{\partial \mathbf{y}_0}(x; x_0, \mathbf{y}_0)$, 则

$$\begin{cases} U'(x; x_0, \mathbf{y}_0) = \frac{\partial \mathbf{f}}{\partial \mathbf{y}}(x, \phi(x; x_0, \mathbf{y}_0))U(x; x_0, \mathbf{y}_0), \\ U(x_0; x_0, \mathbf{y}_0) = I, \end{cases}$$

由 Liouville 恒等式可得

$$\det U(x; x_0, \mathbf{y}_0) = \det U(x_0; x_0, \mathbf{y}_0) e^{\int_{x_0}^x \text{tr}\left(\frac{\partial \mathbf{f}}{\partial \mathbf{y}}(s, \phi(s; x_0, \mathbf{y}_0))\right) ds},$$

因此

$$\det \frac{\partial \phi}{\partial \mathbf{y}_0}(x; x_0, \mathbf{y}_0) = e^{\int_{x_0}^x \text{tr}\left(\frac{\partial \mathbf{f}}{\partial \mathbf{y}}(s, \phi(s; x_0, \mathbf{y}_0))\right) ds}.$$

□